

Executive Summary

Project Title

Product Experimentation & A/B Testing Platform

Project Overview

This project was developed to evaluate whether a newly designed landing page performs better than the existing landing page in terms of user conversions. A/B testing is one of the most widely used product analytics techniques in technology companies such as Google, Amazon, Meta, Microsoft, Airbnb, and Uber to make data-driven product decisions.

The objective of this experiment was to determine if the treatment page (new landing page) significantly improves conversion rates compared to the control page (old landing page).

Business Problem

Organizations frequently introduce new product features, website designs, and user experiences. However, launching changes without validation can negatively impact revenue and customer experience.

The key business question was:

Should the new landing page be rolled out to all users, or should the existing page be retained?

To answer this question, an A/B testing framework was implemented using statistical analysis and experimentation techniques.

Methodology

The project followed a complete experimentation workflow:

1. Data Cleaning and Validation
 - Removed inconsistent records and invalid page assignments.
 - Identified duplicate users and data quality issues.
 - Converted timestamps and created time-based features.
2. Exploratory Data Analysis
 - Analyzed user distribution across control and treatment groups.
 - Calculated conversion rates and conversion lift.
 - Evaluated hourly and daily conversion trends.
3. Statistical Testing
 - Conducted hypothesis testing using a two-proportion Z-test.
 - Calculated confidence intervals.
 - Measured statistical significance using p-values.
4. Business Recommendation
 - Converted analytical findings into actionable product decisions.
 - Evaluated potential business impact of rollout.

Tools & Technologies

- SQL (Data Validation & KPI Analysis)
- Python (Pandas, NumPy, Matplotlib, Statsmodels)
- Statistical Hypothesis Testing
- Power BI
- GitHub

Project Outcome

The experiment successfully measured the performance difference between the control and treatment landing pages and provided a data-driven framework for making rollout decisions based on statistical evidence rather than assumptions.

Final Insights & Business Recommendations

Key Insights

User Conversion Analysis

The analysis revealed measurable differences in conversion behavior between users exposed to the control page and those exposed to the treatment page.

Key metrics analyzed included:

- Total Users
- Total Conversions
- Overall Conversion Rate
- Control Conversion Rate
- Treatment Conversion Rate
- Conversion Lift

Time-Based Insights

User conversion behavior varied across different hours and days.

Findings included:

- Identification of peak conversion hours.
- Detection of low-engagement periods.
- Analysis of user activity patterns throughout the experiment.

These insights can help optimize marketing campaigns and product launches during high-performing time windows.

Experiment Performance

The treatment page demonstrated improved conversion performance compared to the control page.

The experiment quantified:

- Absolute Conversion Difference
- Relative Conversion Lift
- Statistical Significance
- Confidence Intervals

This ensured that observed improvements were evaluated scientifically rather than through random fluctuations.

Statistical Findings

A two-proportion Z-test was conducted to validate the experiment results.

The analysis measured:

- Z-Score
- P-Value
- Confidence Interval

These metrics were used to determine whether the observed performance difference was statistically significant and suitable for business decision-making.

Business Impact

This project demonstrates how experimentation can reduce decision-making risk and improve product performance.

Potential benefits include:

- Increased customer conversions
- Improved user experience
- Better product adoption
- Revenue growth opportunities
- Reduced risk during product launches

Final Recommendation

Based on the experiment results and statistical evaluation, the treatment landing page should be considered for rollout if it demonstrates a statistically significant improvement over the control page.

Organizations should continue monitoring post-rollout performance to ensure that improvements remain stable over time.

Conclusion

This project showcases a complete end-to-end Product Analytics workflow, including data validation, exploratory analysis, experimentation, statistical testing, visualization, and business recommendation generation.

The framework mirrors real-world experimentation processes used by leading technology companies and demonstrates practical skills required for Product Analyst and Data Analyst roles.